Maven Project with Guava and Commons-IO Dependencies

Create a Maven Project Using Terminal by following command:

mvn archetype: generate -DgroupId=com.example - DartifactId=MyMavenGuavaApp - DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
vboxuser@prathwish03:~/devops$ mvn archetype:generate   -DgroupId=com.example   -DartifactId=MyMavenGuavaApp   -Darc
etypeArtifactId=maven-archetype-quickstart   -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMavenGuavaApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.354 s
[INFO] Finished at: 2026-03-21T09:56:06+05:30
```

Now Navigate to the Project Directory using **cd MyMavenGuavaApp**

Edit the pom.xml File and add the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/maven- v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId><artifactId>MyMavenGuavaApp</artifactId>
      <packaging>jar</packaging>
      <version>1.0-SNAPSHOT</version>
      <name>MyMavenGuavaApp</name>
      <url>http://maven.apache.org</url>
      <dependencies>
        <dependency>
          <groupId>junit</groupId>
          <artifactId>junit</artifactId>
          <version>3.8.1</version>
          <scope>test</scope>
        </dependency>
```

```
   <!-- https://mvnrepository.com/artifact/com.google.guava/guava -->
<dependency>
   <groupId>com.google.guava</groupId>
   <artifactId>guava</artifactId>
   <version>33.4.0-jre</version>
</dependency>

<!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
<dependency>
   <groupId>commons-io</groupId>
   <artifactId>commons-io</artifactId>
   <version>2.18.0</version>
</dependency>

  </dependencies>

  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>

<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.1.0</version>
        <configuration>
```

```
            <archive>
              <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
              </manifestEntries>
            </archive>
          </configuration>
        </plugin>
  </plugins>
  </build>
  </project>
```

Modify App.java

Got to App.java using **cd src/main/java/com/example/** and **gedit App.java** to type the file with following code:

```
package com.example;

/**
 * Hello world!
 *
 */
// Import FileUtils class from Apache Commons IO library
// This is used to perform file operations like
copying files import
org.apache.commons.io.FileUtils;

// Import File class to represent file and directory
pathnames import java.io.File;

// Import IOException to handle input-output
errors import java.io.IOException;

// Import ImmutableList from Google Guava library
// ImmutableList creates a list that cannot be modified after creation
```

```
//import com.google.common.collect.ImmutableList;

// Define the main class
public class App
{
   // Main method - program execution starts from
   here public static void main( String[] args )
   {
      // Create an immutable list of strings (fruits)
      // This list cannot be changed (no add/remove allowed)
      ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");

      // Print the list of fruits to the console
      System.out.println(fruits);

      // Create a File object representing the
      source file File sourceFile = new
      File("source.txt");

      // Create a File object representing the
      destination file File destFile = new
      File("destination.txt");

      try {
         // Copy the contents of source.txt to destination.txt
         // Using Apache Commons IO FileUtils class
         FileUtils.copyFile(sourceFile, destFile);

         // Print success message if file is copied
         System.out.println("File copied successfully!");
      }
      catch (IOException e) {
```

```
            // If any error occurs during file copying,
            // this block will handle the exception
            // Print error message
            System.err.println("Error occurred while copying file: " + e.getMessage());
        }
    }
}
```

Build the Project using the command **mvn clean install** command

```
vboxuser@prathwish03:~/devops/MyMavenGuavaApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenGuavaApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vboxuser/devops/MyMavenGuavaApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vboxuser/devops/MyMavenGuavaApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -------------------------------------------------------
```

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.059 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO] Building jar: /home/vboxuser/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/vboxuser/devops/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /home/vboxuser/.m2/re
App/1.0-SNAPSHOT/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/vboxuser/devops/MyMavenGuavaApp/pom.xml to /home/vboxuser/.m2/repository/com/example/MyMavenGuav
-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.495 s
[INFO] Finished at: 2026-03-21T10:02:34+05:30
[INFO] ------------------------------------------------------------------------
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

If java-jar command give an error which is most probable, use the following command mvn exec:java -Dexec.mainClass=”com.example.App”

```
vboxuser@prathwish03:~/devops/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass=com.example.App
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
Error occurred while copying file: Source 'source.txt' does not exist
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.704 s
[INFO] Finished at: 2026-03-21T11:25:10+05:30
[INFO] ------------------------------------------------------------------------
```

This command gives error asError while copying file:Source ‘source.txt’ does not exist

Now, Create a Sample File to copy the content to destination file

```
Open    source.txt
        ~/report/MyMavenGuavaApp    Save
1 Copying contents of this file to the destination file
```

Now Run the command **mvn exec:java -Dexec.mainClass=”com.example.App”** again

```
vboxuser@prathwish03:~/devops/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass=com.example.App
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenGuavaApp >--------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.623 s
[INFO] Finished at: 2026-03-21T11:28:19+05:30
[INFO] ------------------------------------------------------------------------
```

The output shows file copied successfully.

We can see newly created **destination.txt** using gedit destination.txt if you are still in Project’s directory

```
vboxuser@prathwish03:~/devops/MyMavenGuavaApp$ ls
destination.txt  pom.xml  source.txt  src  target
vboxuser@prathwish03:~/devops/MyMavenGuavaApp$
```

File content is same as the content in source file.

Guava pipeline

Step 1: Create new file named "**Jenkinsfile**" inside MyMavenGuavaApp folder and push into git

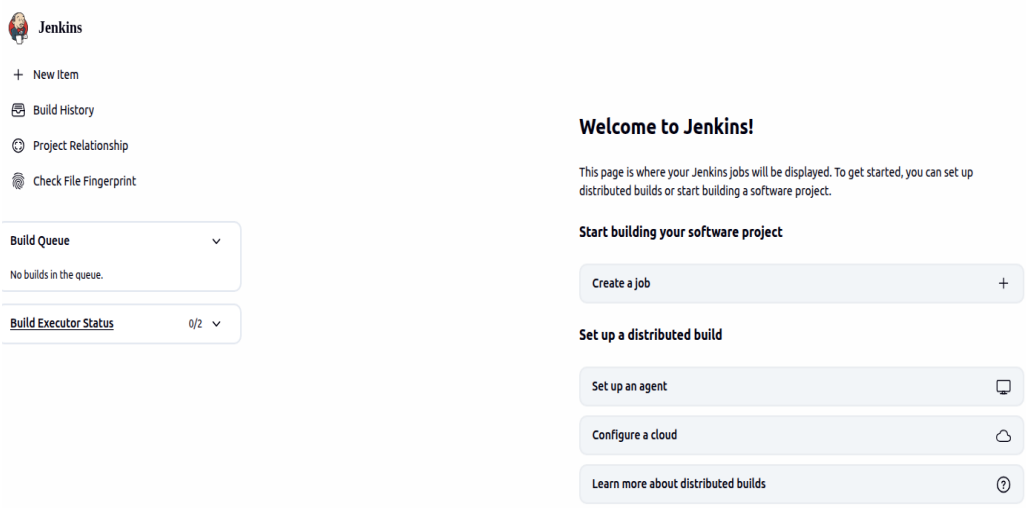
Jenkinsfile:

```
pipeline {
   agent any  // Use any available agent

   tools {
      maven 'Maven'   // Ensure this matches Jenkins configuration
      jdk 'JDK'
   }

   stages {
      stage('Checkout') {
         steps {
            git branch: 'main', url:
'https://github.com/PrathwishShetty03/MyMavenGuavaApp.git'
         }
      }
      stage('Build') {
         steps {
            sh 'mvn clean compile'
         }
      }
      stage('Test') {
         steps {
            sh 'mvn test'
         }
      }
      stage('Package') {
         steps {
            sh 'mvn package'
         }
      }
      stage('Run Application') {
         steps {
            sh 'java -jar target/MyMavenGuavaApp-1.0-SNAPSHOT.jar'
         }
```

```
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

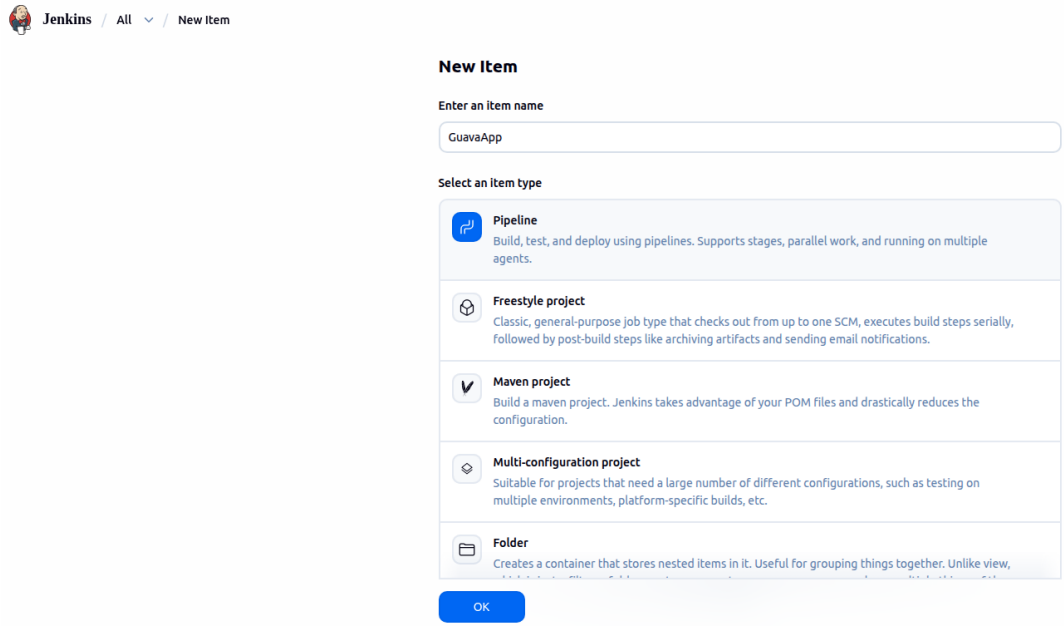
Step 2:Open Jenkins Dashboard

- Open Jenkins in browser
- Click on New Item



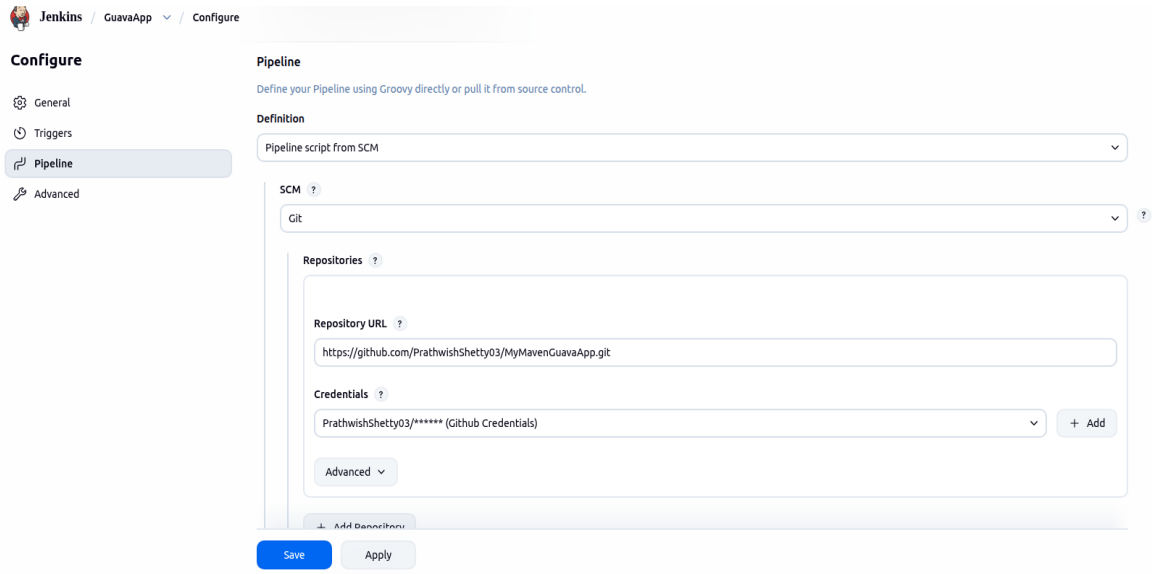
Step 3: Create Pipeline Job

- Enter name: **GuavaApp**
- Select **Pipeline**
- Click **OK**



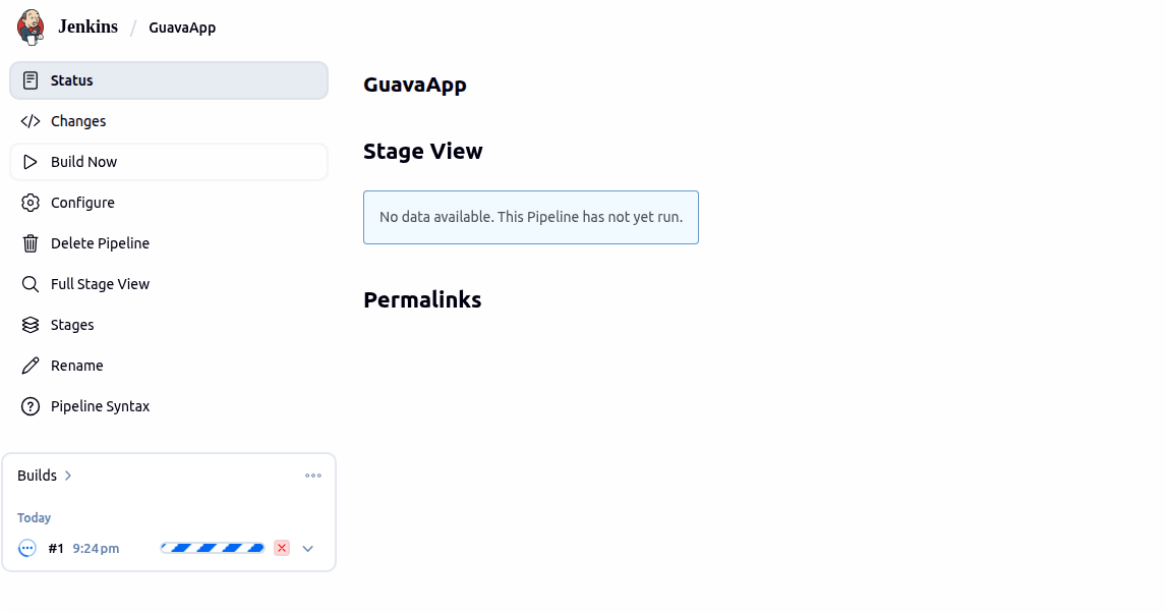
Step 4: Configure Pipeline with GitHub

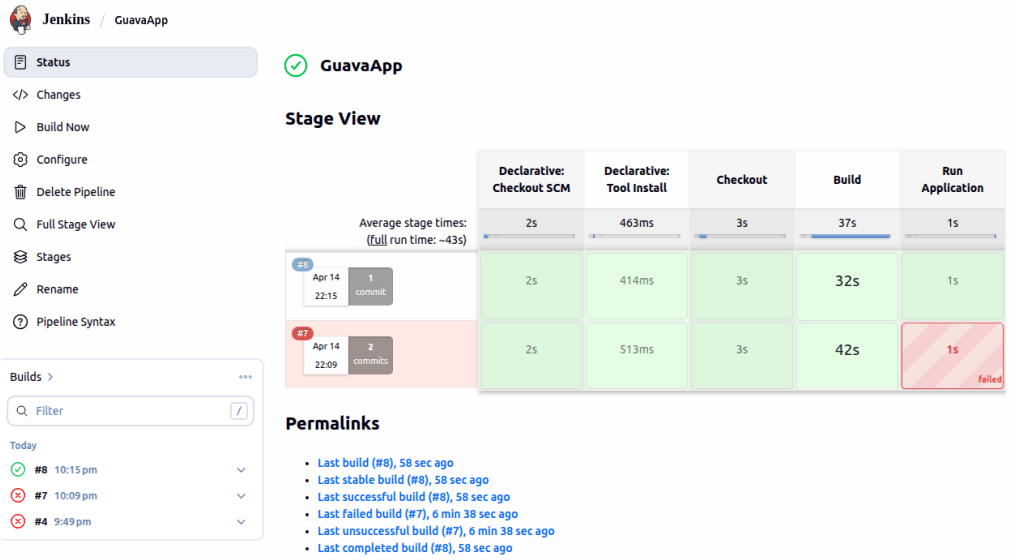
- Select **Pipeline script from SCM**
- SCM → Git
- Enter repository URL
- Add credentials
- Click **Save**



Step 5: Run the Pipeline

- Click **Build Now**



Step 6: Look for the status of project

- Check Stage View
- View Build History